

C2_02 Module Test - Protected Review

RC4

Open only after a saved genuine attempt. Diagnose the weak competency, close this review, then prove repair on a changed query/fixture.

ASQL1

Aggregate/validate the many side to intended output grain or use the correct grain-specific measure; prove row/distinct-key parity instead of hiding fan-out with DISTINCT.

ASQL2

Unsafe when it can return multiple rows/columns where one value is expected, or when duplicates/NULL semantics make the comparison ambiguous.

ASQL3

Named stages expose grain and controls so each transformation can be inspected independently; readability/validation is the goal, not forcing physical execution order.

ASQL4

GROUP BY collapses to one row/group; a window calculation preserves row grain and adds a value computed over a partition/order/frame.

ASQL5

State tie behavior and deterministic secondary ordering. ROW_NUMBER forces unique positions; RANK/DENSE_RANK preserve ties differently.

ASQL6

LAG returns the previous existing row, not necessarily the previous calendar period. Build/check the period series or detect gaps.

ASQL7

Define eligible population first, then numerator subset; denominator must not drift because of joins, NULLs or hidden status exclusions.

ASQL8

Half-open intervals are precision-safe and non-overlapping: include start, exclude next boundary. 23:59:59 can miss higher-precision timestamps.

ASQL9

Duplicates, orphan foreign keys, null/invalid fields, impossible values/ranges, row-count and monetary reconciliation are common minimum checks.

ASQL10

Compare rows/distinct keys/control totals immediately before and after the first suspect stage; isolate first divergence.

ASQL11

Planner route: scans, join methods/order, estimates, index usage and costs. Runtime alone does not explain why or whether logic is correct.

ASQL12

Clear inputs/grain/assumptions, named stages, independent controls, explicit period rules, safe performance evidence, comments/handoff and a fresh rerun path.

Repair routing

Weak evidence	Repair route
grain / join	ASQL1 + remediation grain/join; changed join fixture
architecture / window / time	ASQL3-ASQL8 + window/time remediation; changed period fixture
validation / performance	ASQL9-ASQL12 + validation/performance remediation; changed control query

This diagnostic remains non-gating. Gate A eligibility is unchanged.